Applied Data Mining

— Week 8—

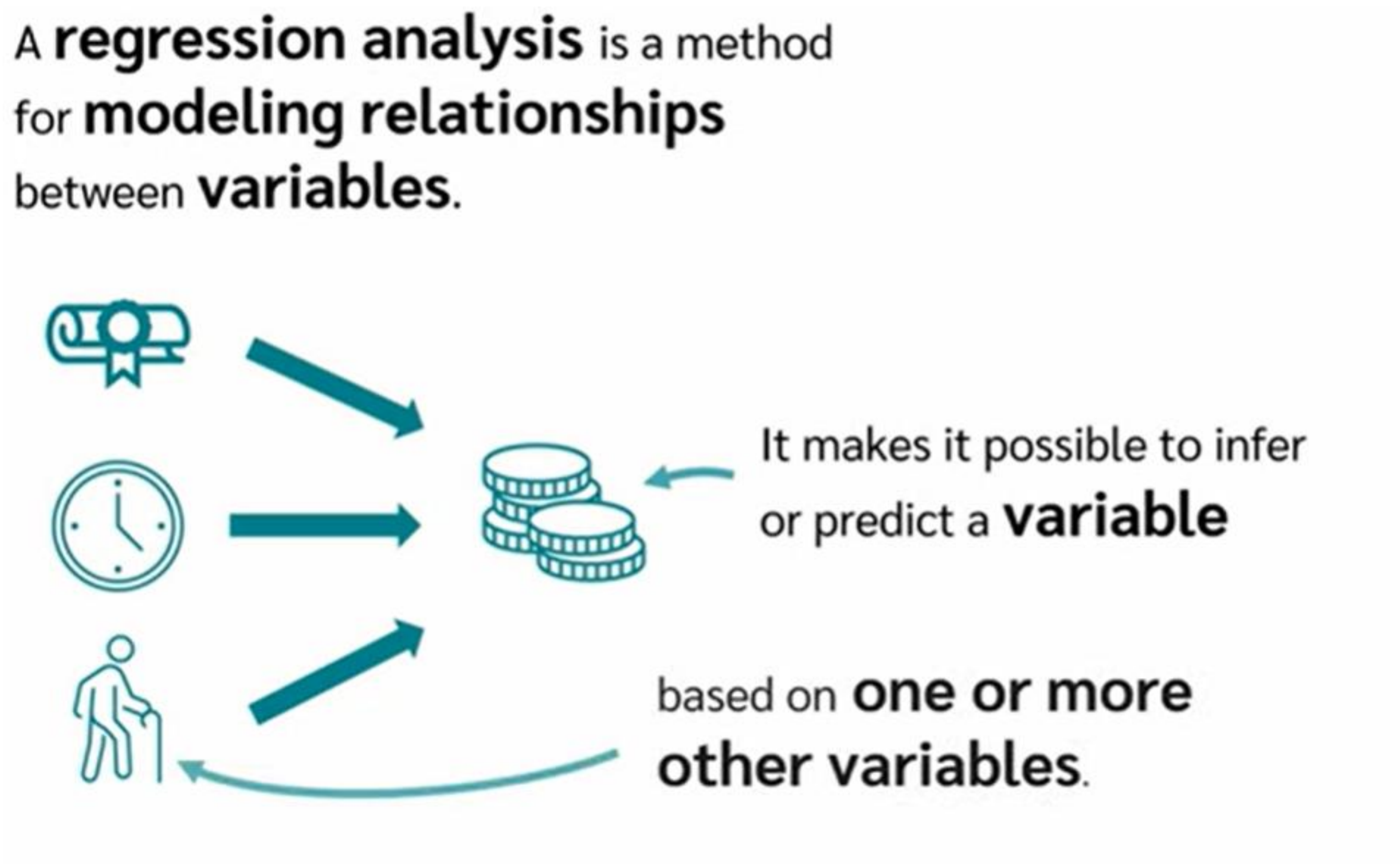
Spring 2026

Dr. Hashim Safdar

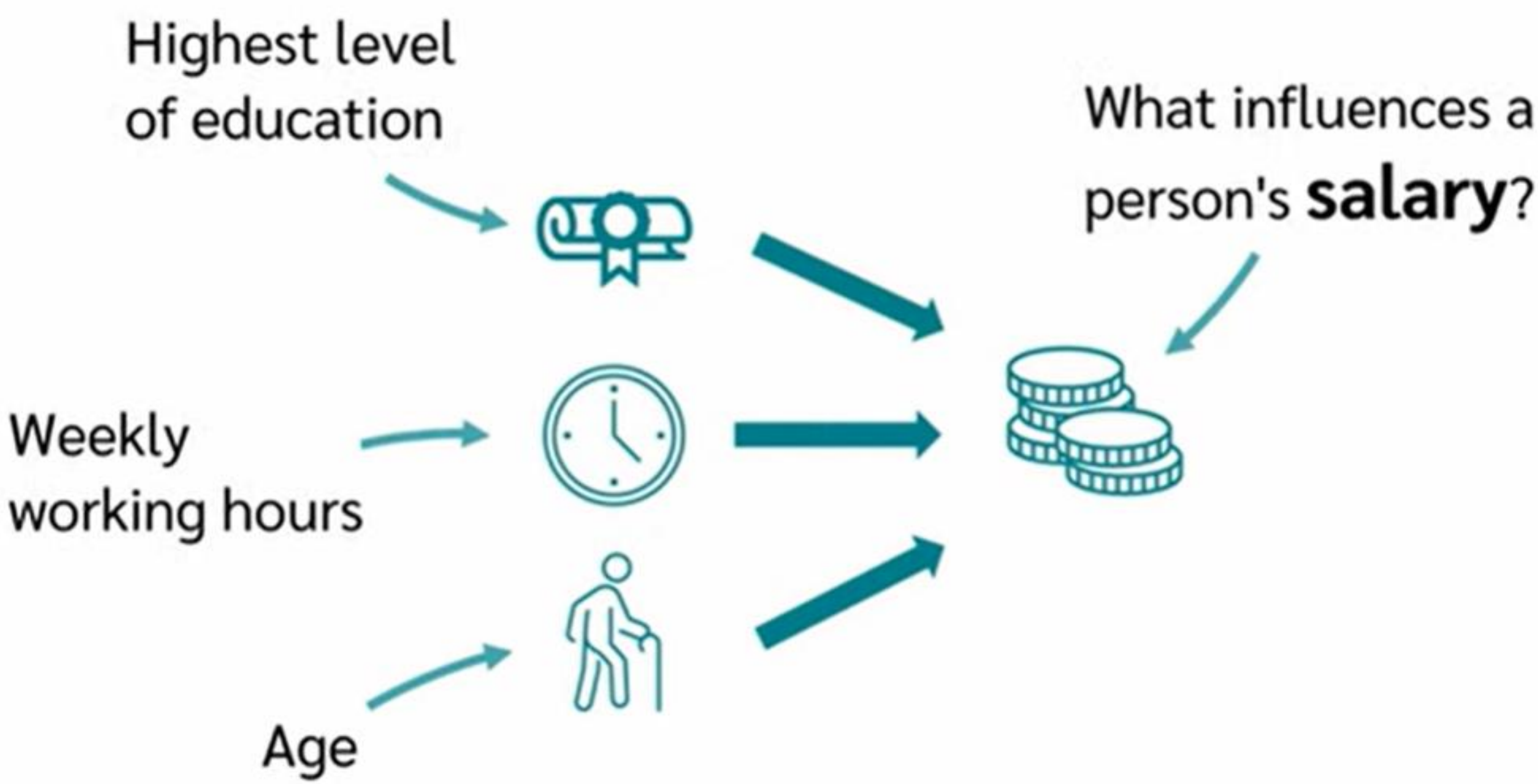
Regression Models

Introduction, Smoothing,
Classification & Prediction

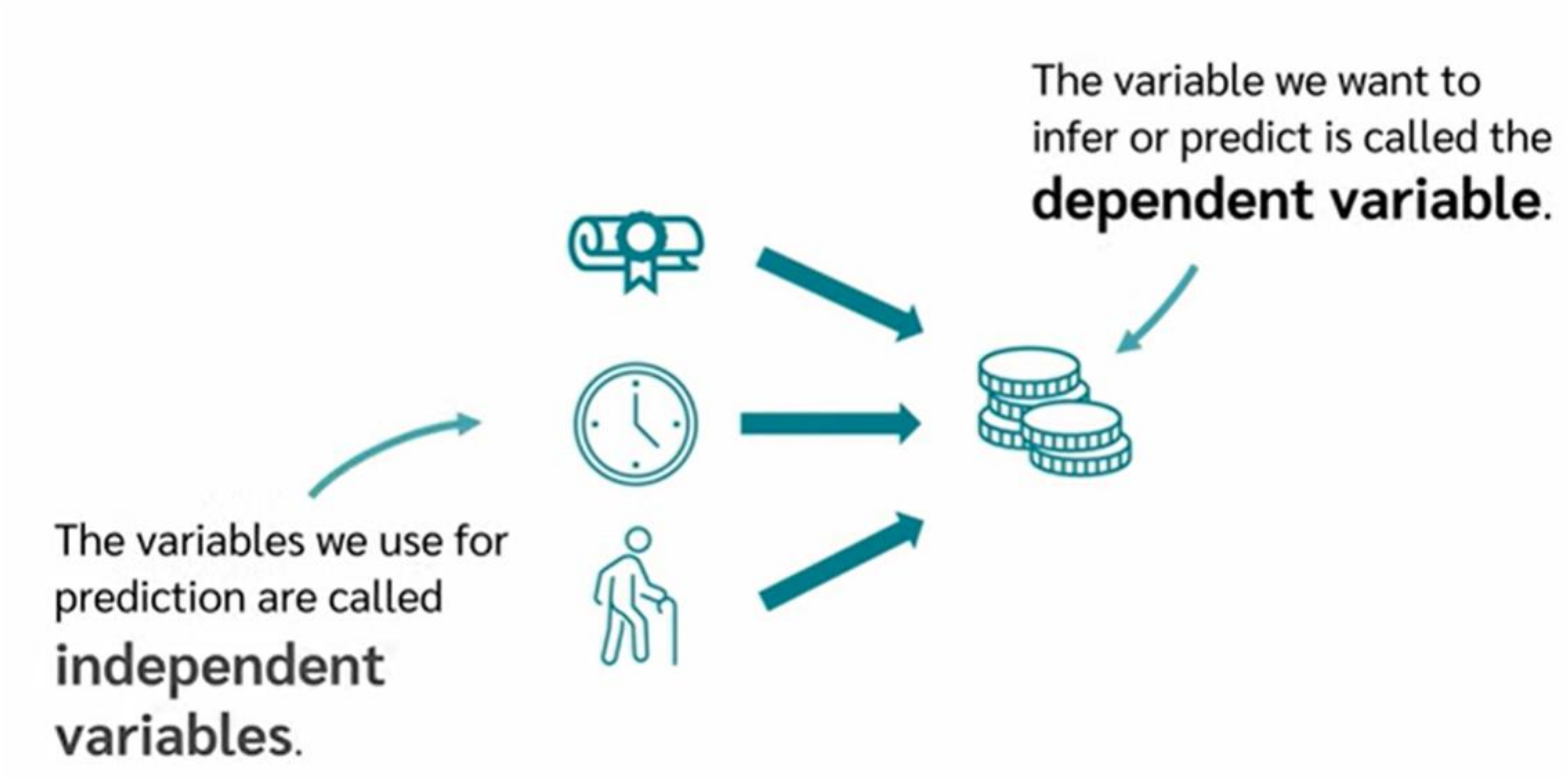
What is Regression Analysis?



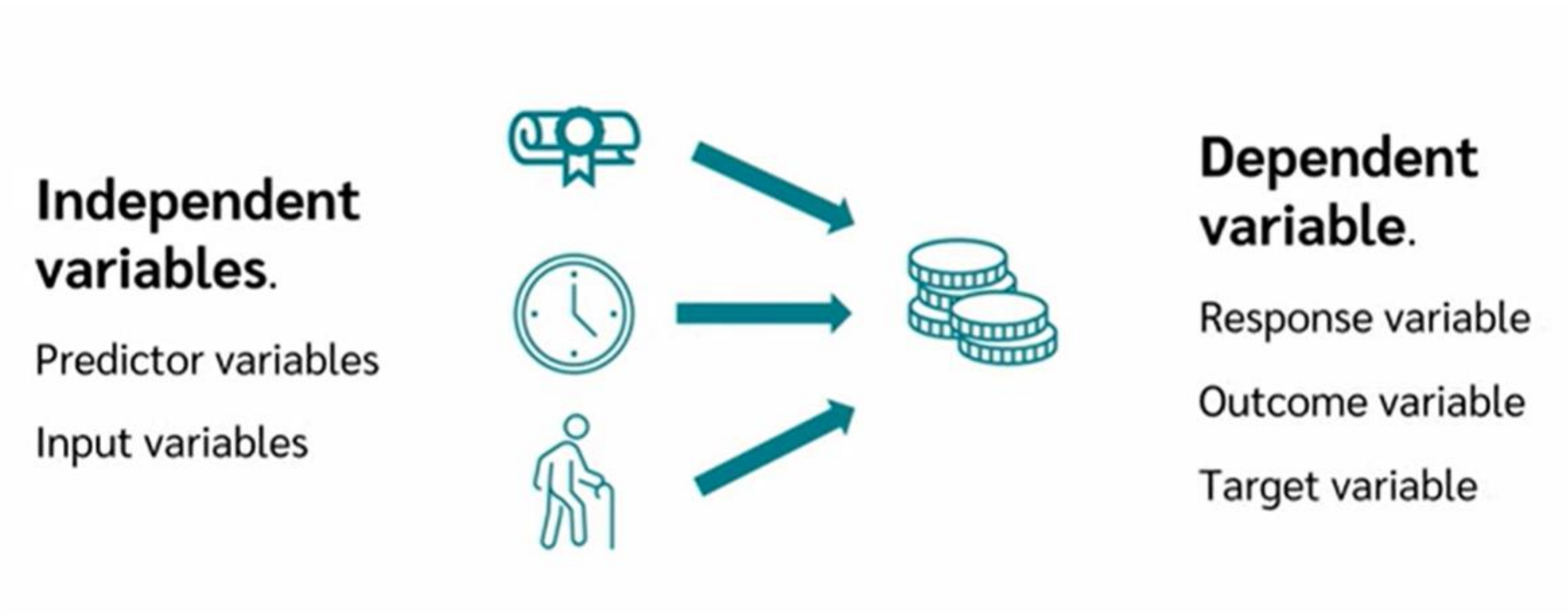
What is Regression Analysis?



What is Regression Analysis?



What is Regression Analysis?



Regression = relationship + prediction + modeling

When to use Regression Analysis?

Regression analysis can be used to achieve **two goals**.

1. **Measure the influence of one or more variables on another variable.**
2. **Prediction of one variable by one or more other variables.**

When to use Regression Analysis?

1 **Measure the influence of one or more variables on another variable.**

What influences children's ability to concentrate?

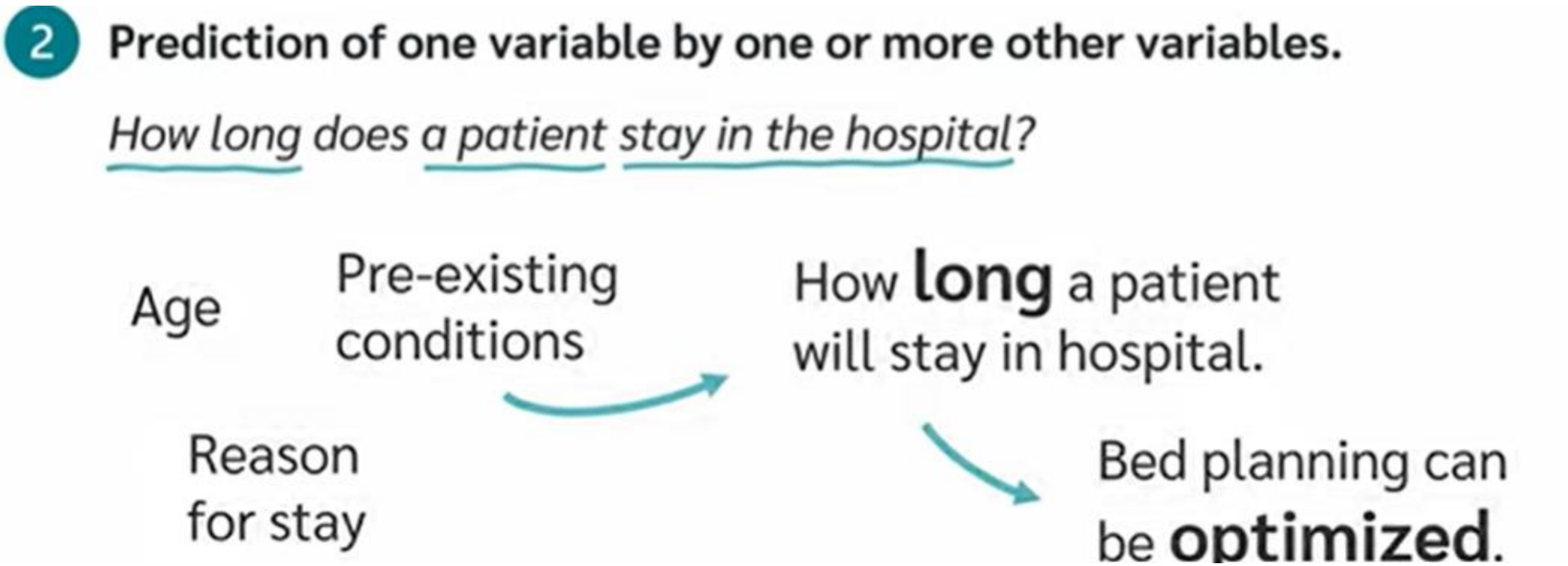
Do the educational level of parents and place of residence have an influence on the future educational level of children?

This area is **research-based**

and has many applications in the

social and **economic sciences**.

When to use Regression Analysis?



When to use Regression Analysis?

2 Prediction of one variable by one or more other variables.

How long does a patient stay in the hospital?

Age

Pre-existing conditions

Reason for stay

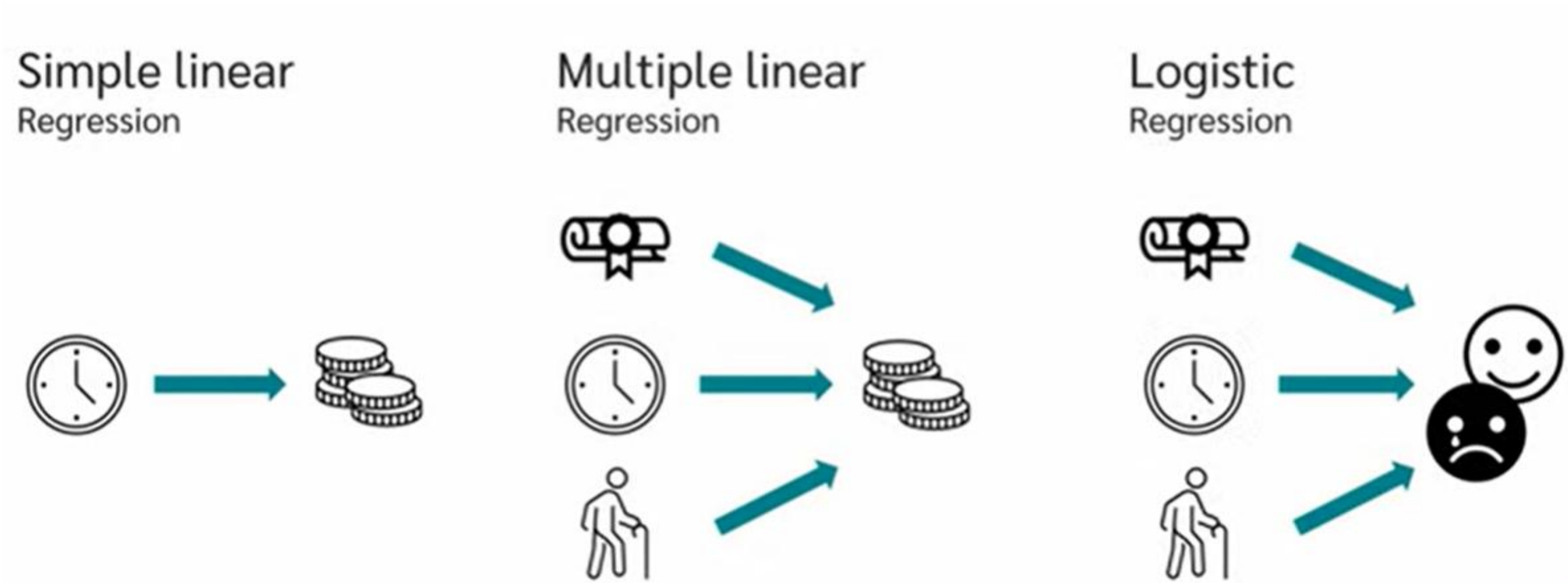
How **long** a patient will stay in hospital.

Bed planning can be **optimized**.

What product is a person most likely to buy from an online store?

Highly **application-oriented**, focusing on making **predictions** to enhance **efficiency**.

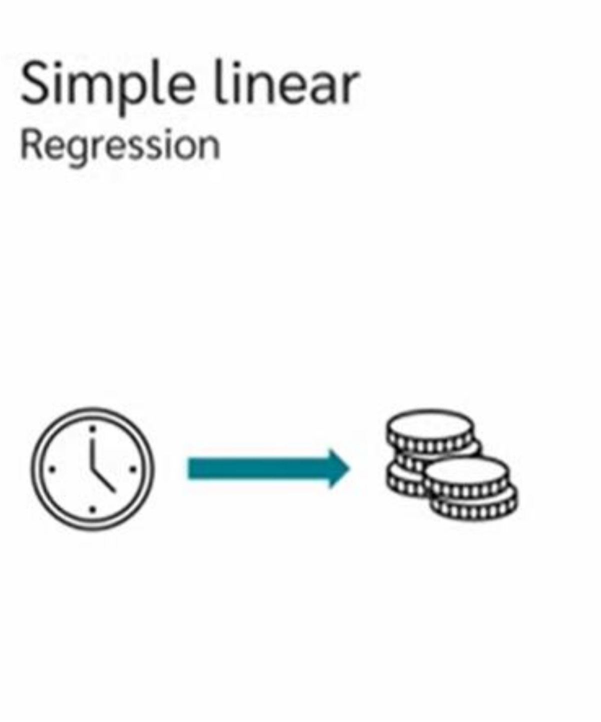
Types of Regression Analysis?



Types of Regression Analysis?

Linear regression is a statistical method used to **predict the value of one variable (dependent variable Y) based on another variable (independent variable X)**.

$Y = b_0 + b_1 X$



Types of Regression Analysis?

Multiple regression is an extension of simple linear regression.

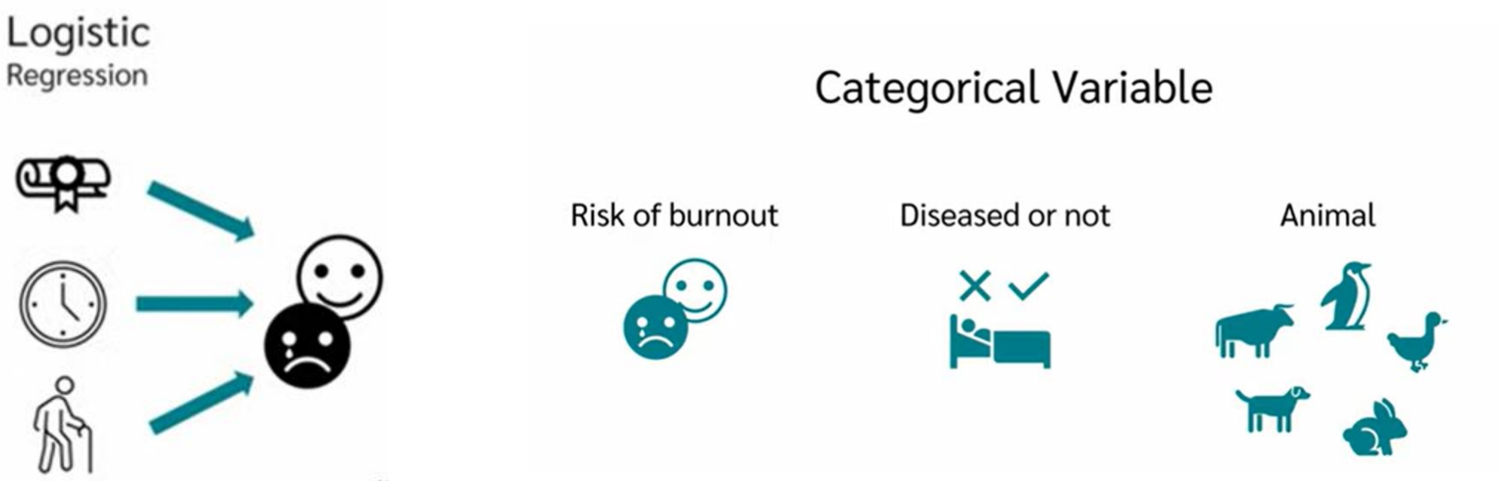
Instead of **predicting Y from one independent variable, you predict it from two or more independent variables.**

Types of Regression Analysis?

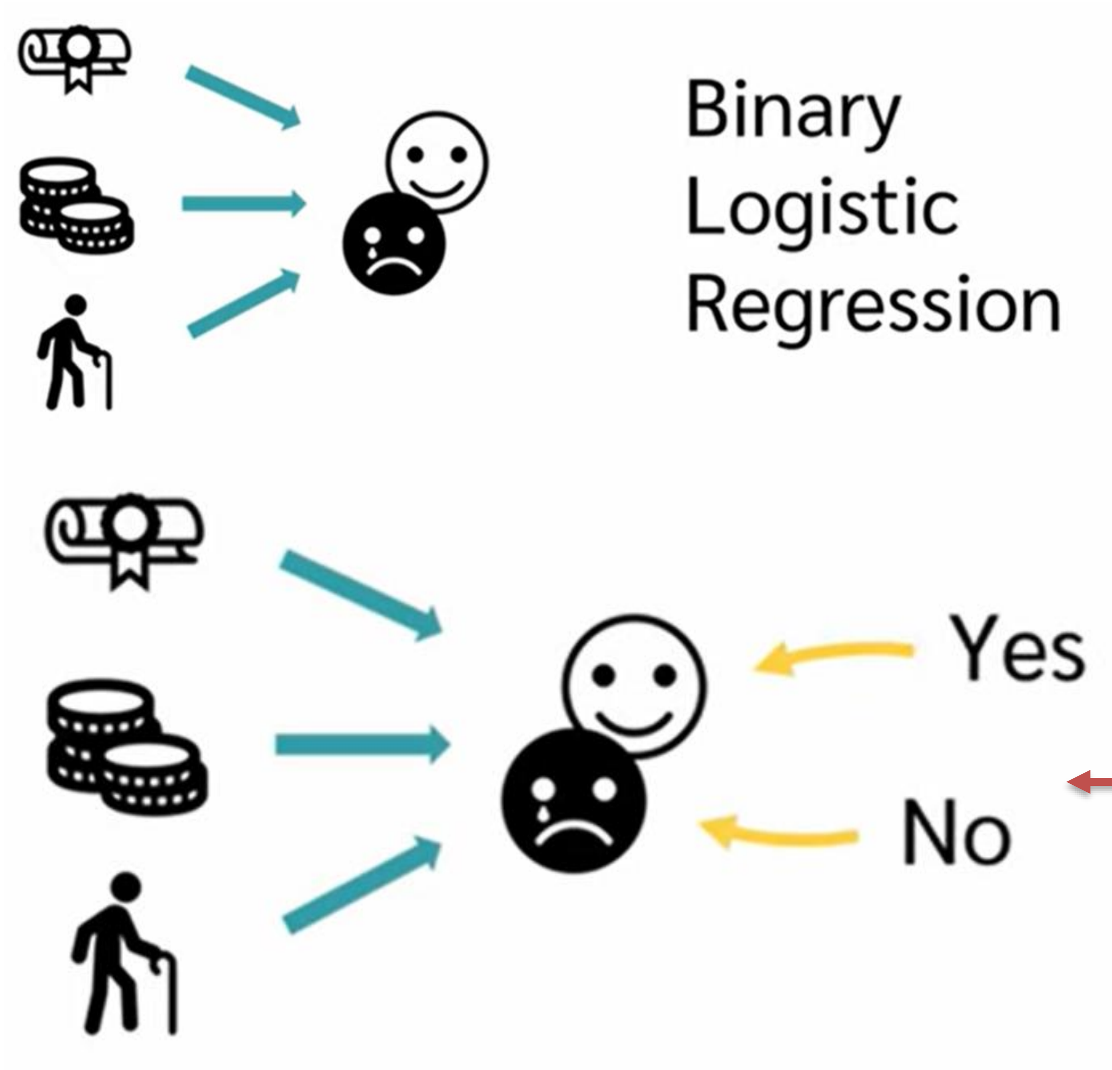
Logistic regression is used when your **dependent variable (Y) is categorical**, usually **binary** (0 or 1, Yes or No, True or False).

•Unlike linear regression, it **predicts probabilities**, not exact numbers, and maps them to **0 or 1**.

Example: Predict whether a student **passes (1) or fails (0)** based on study hours.

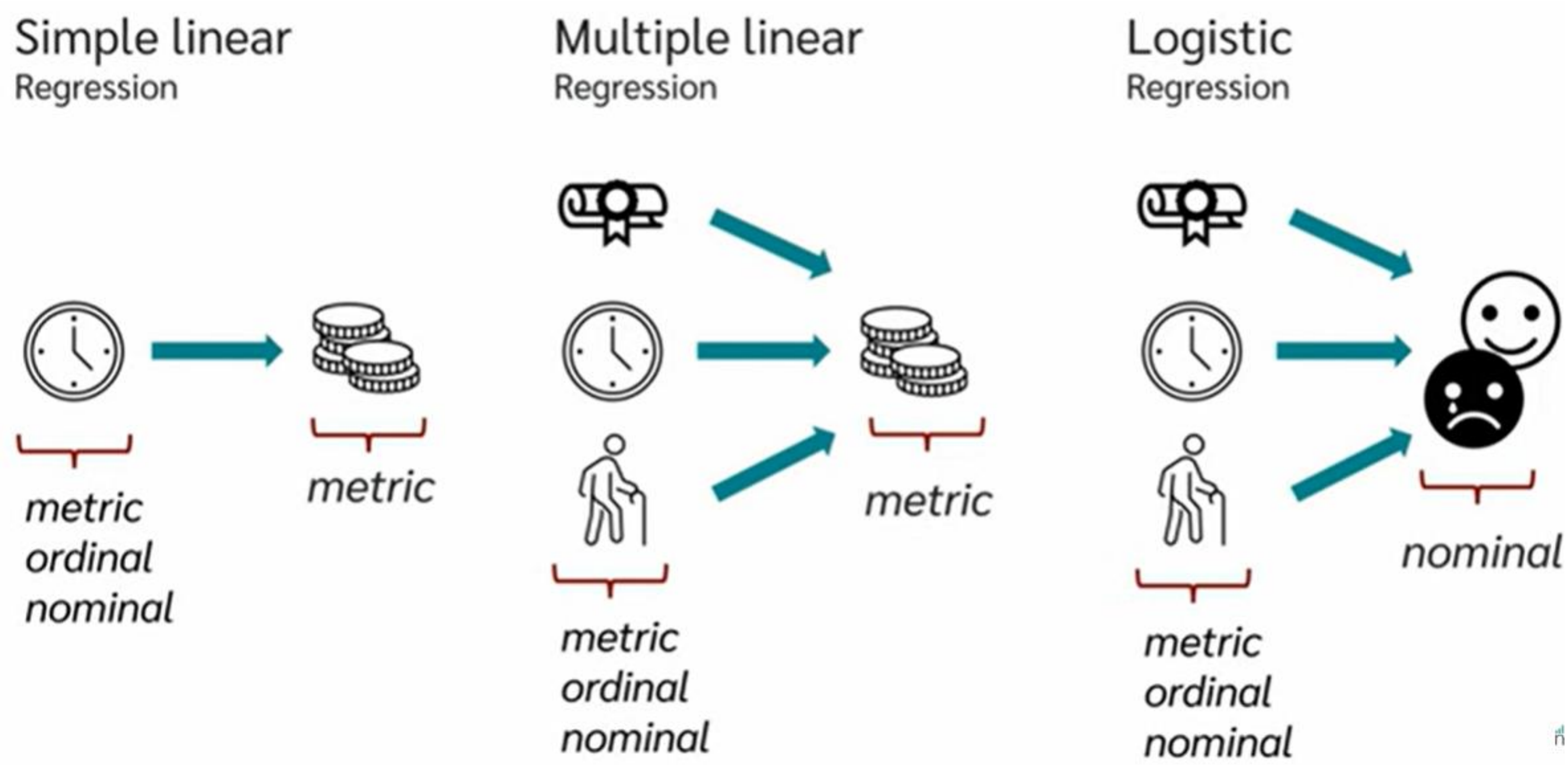


Types of Regression Analysis?



Two Possible Values

Types of Regression Analysis?



Types of Regression Analysis?

Simple linear regression

Does the **weekly working time** have an influence on the **hourly wage** of employees?

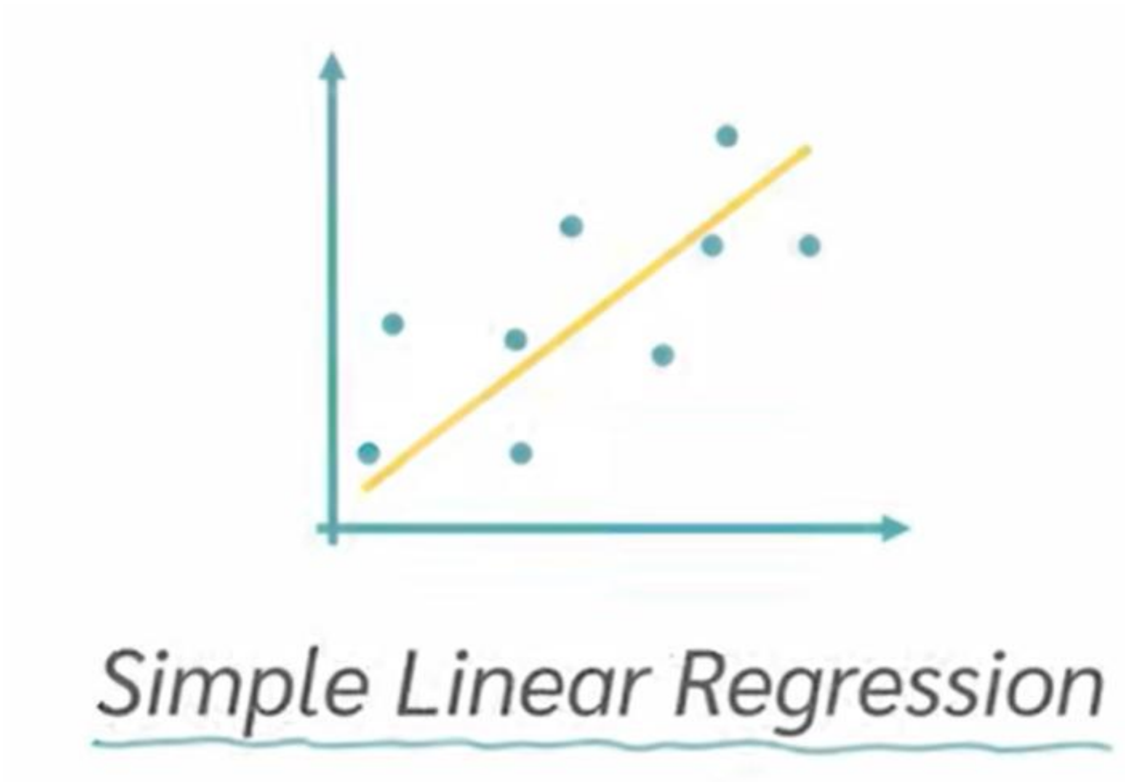
Multiple lineare regression

Do the **weekly working time** and the **age of employees** have an influence on their **hourly wage**?

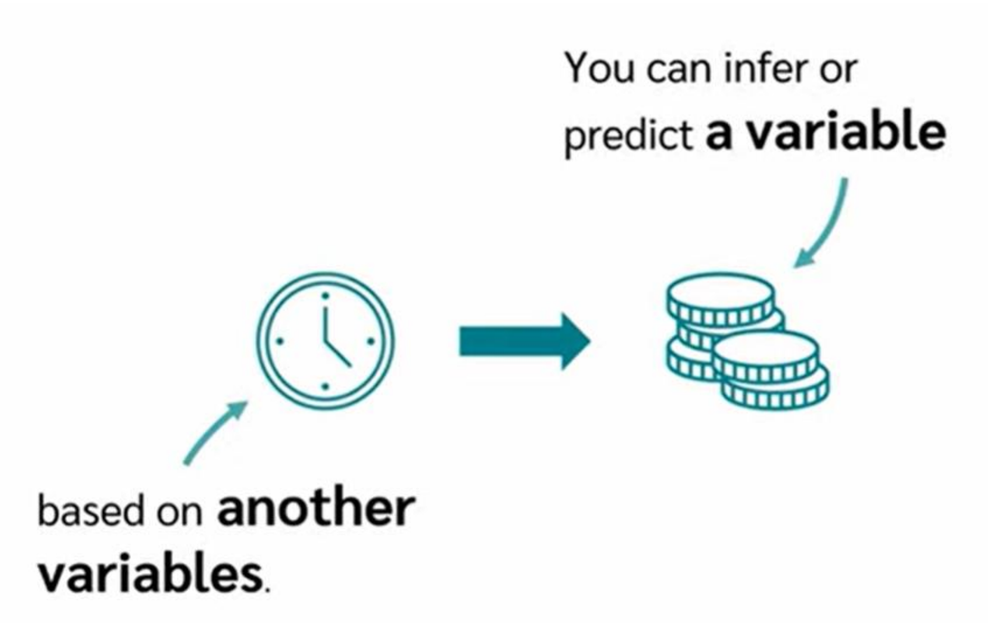
Logistic regression

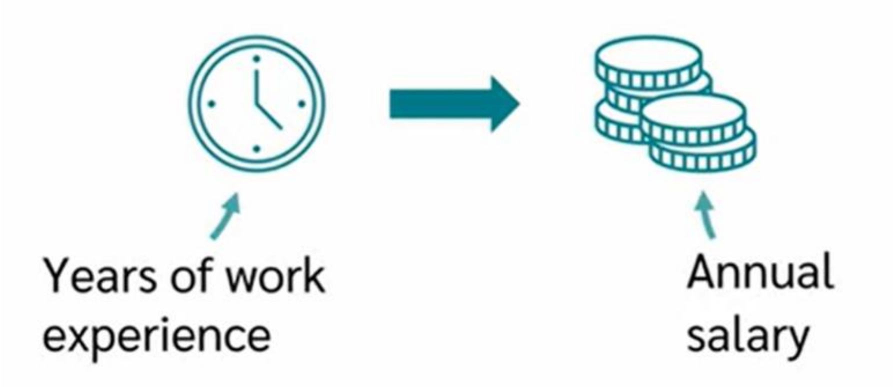
Do the **weekly working time** and the **age** of employees have an influence on the probability that they are **at risk of burnout**?

Simple Linear Regression

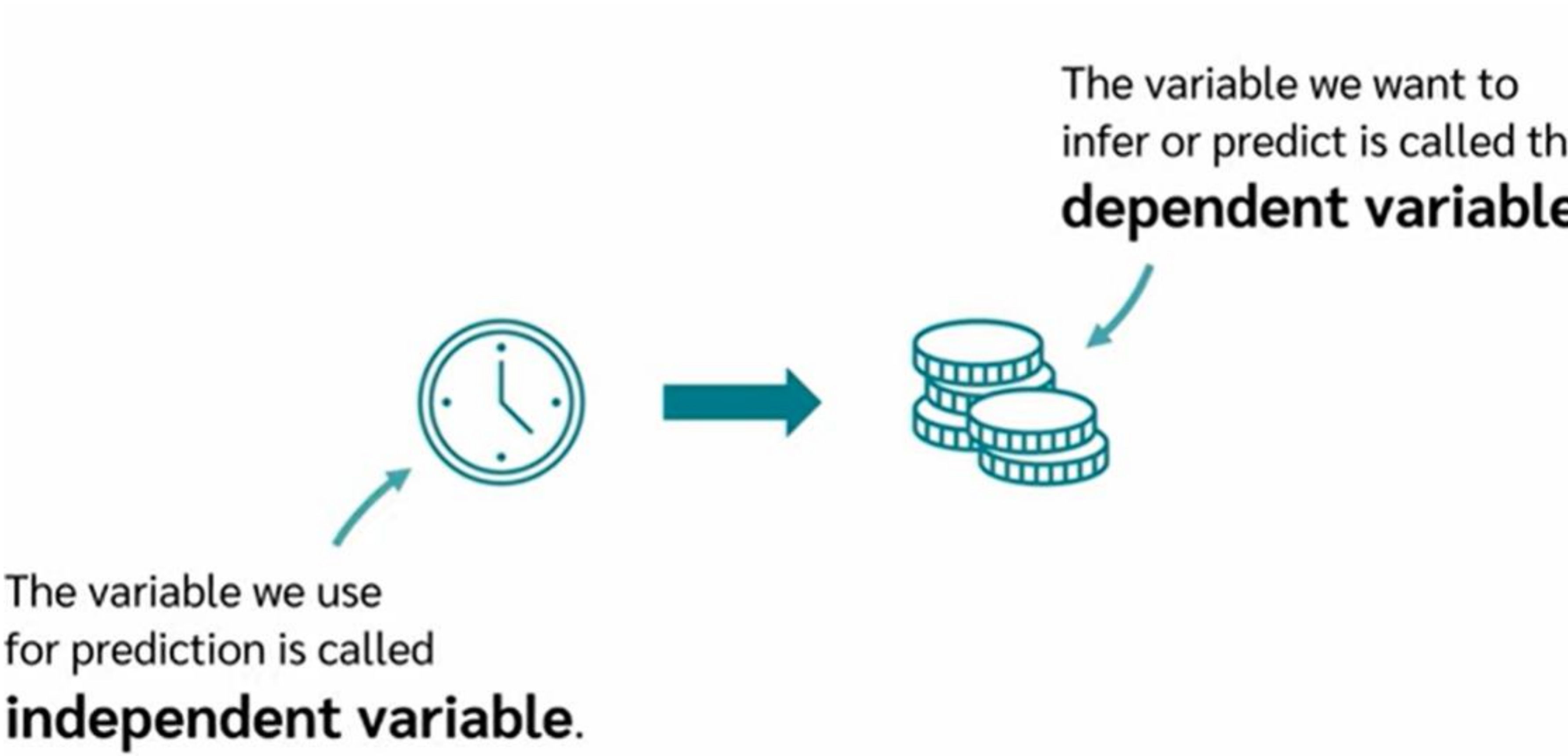


Simple Linear Regression





Simple Linear Regression



Simple Linear Regression

How to Calculate Simple Linear Regression?

First of all, we need **data**.

House Size	House Price
1852	316000
1975	277000
1176	155000
1550	253000
1458	211000
2689	329000
2259	317000
2763	360000
1325	204000
1992	250000

Regression Model $y = bx + a$

How to Calculate Simple Linear Regression?

House Size	House Price
1852	316000
1975	277000
1176	155000
1550	253000
1458	211000
2689	329000
2259	317000
2763	360000
1325	204000
1992	250000

$$y = bx + a$$

House Price *House Size*

How to Calculate Simple Linear Regression?

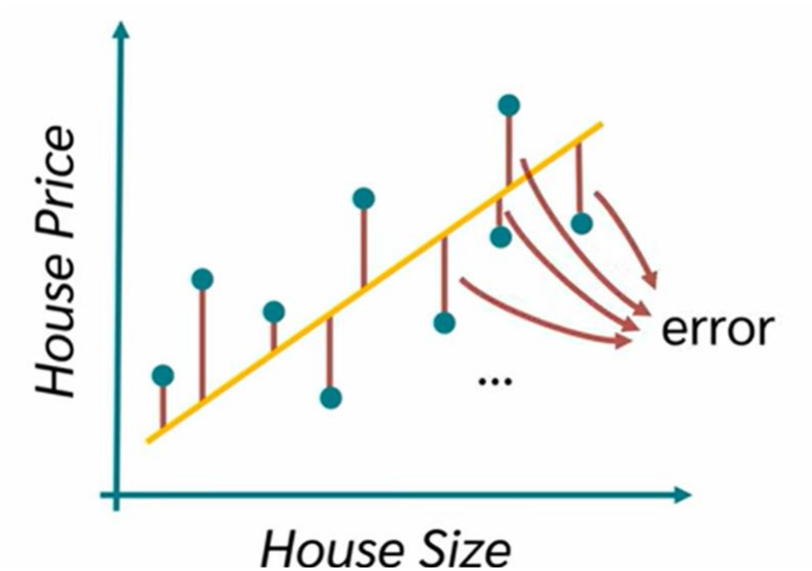
House Size	House Price
1852	316000
1975	277000
1176	155000
1550	253000
1458	211000
2689	329000
2259	317000
2763	360000
1325	204000
1992	250000

We want to use our **data** to determine the coefficient **b** and **a**.

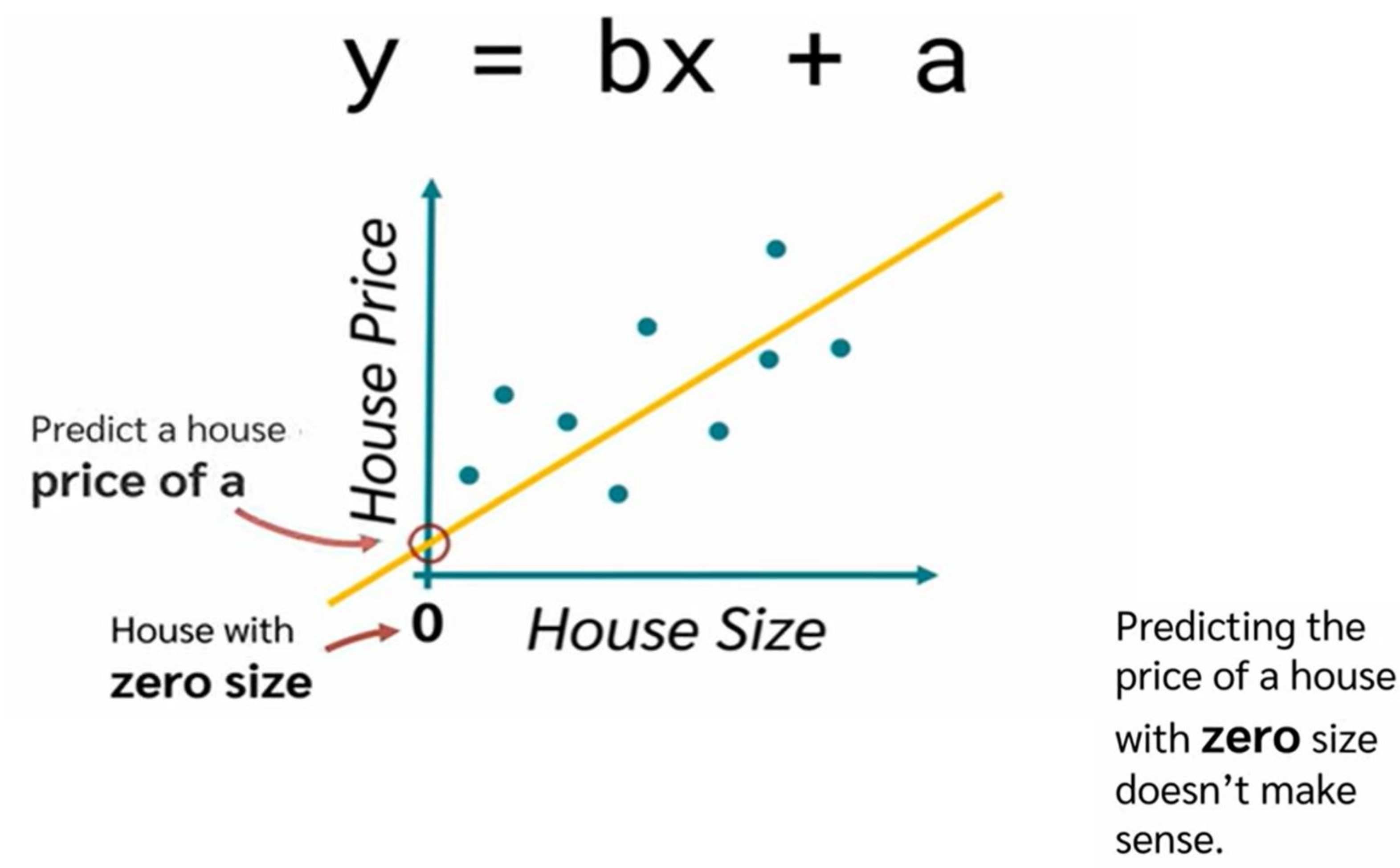
$$y = bx + a$$

How to Calculate Simple Linear Regression?

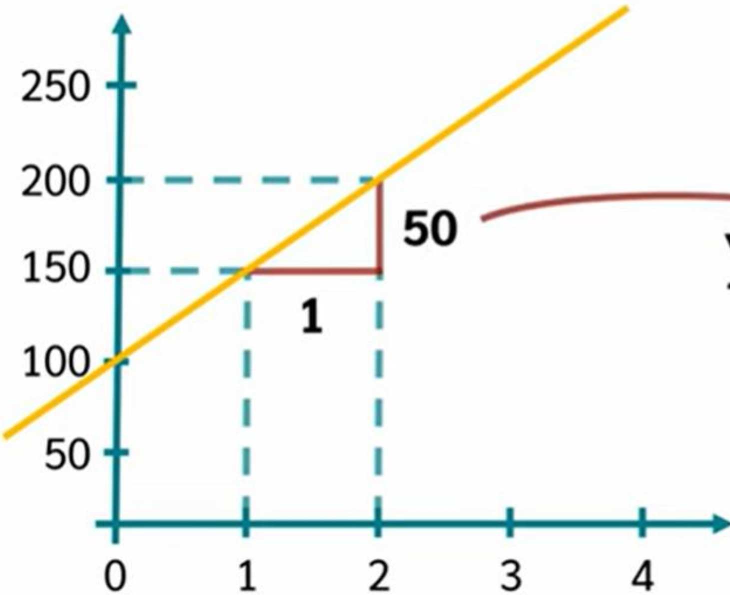
We have to draw (find) a regression line to minimize the error.



How to Calculate Simple Linear Regression?



How to Calculate Simple Linear Regression?



$$y = bx + 100$$

$$\frac{\Delta y}{\Delta x} = \frac{\delta y}{\delta x} = \frac{50}{1} = 50$$

How to Calculate Simple Linear Regression?

$$y = \underline{b}x + \underline{a}$$

$$b = r\frac{s_y}{s_x}$$

House Size	House Price
1852	316000
1975	277000
1176	155000
1550	253000
1458	211000
2689	329000
2259	317000
2763	360000
1325	204000
1992	250000

- "r" is correlation co-efficient of "y" and "x".
- Sx and Sy are standard deviation.

$$b = \underset{0.92}{r}\frac{s_y}{s_x}$$

$$\underset{108.35}{b} = \underset{0.92}{r}\frac{s_y \leftarrow 61341.34}{s_x \leftarrow 518.17}$$

How to Calculate Simple Linear Regression?

$$y = bx + a$$

$$a = \bar{y} - b \cdot \bar{x}$$

267200 108.35 1903.9

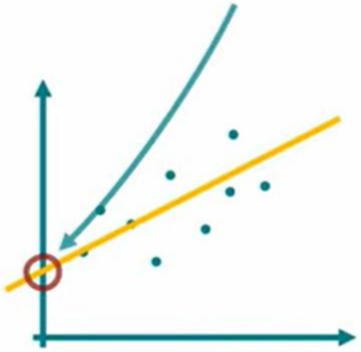
House Size	House Price
1852	316000
1975	277000
1176	155000
1550	253000
1458	211000
2689	329000
2259	317000
2763	360000
1325	204000
1992	250000

$\bar{y}$ and $\bar{x}$ are mean values of house price and house size.

How to Calculate Simple Linear Regression?

$$y = 108.35\,x + 60919.44$$

$$y = 108.35 \cdot 0 + 60919.44 = 60919.44$$



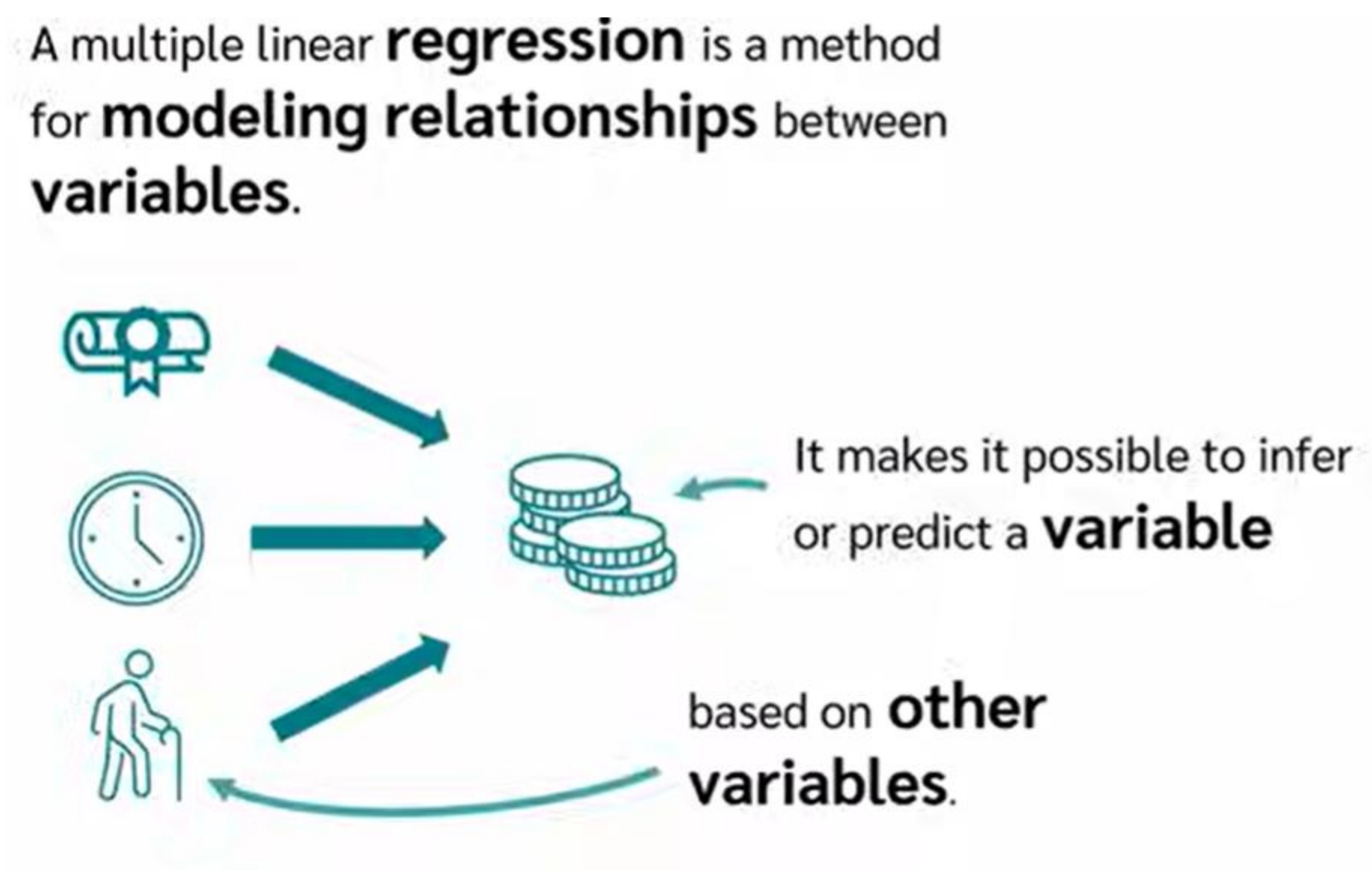
$$y = 108.35 \cdot \mathbf{1} + 60919.44 = 61027.79$$

$$y = 108.35 \cdot \mathbf{2} + 60919.44 = 61136.14$$

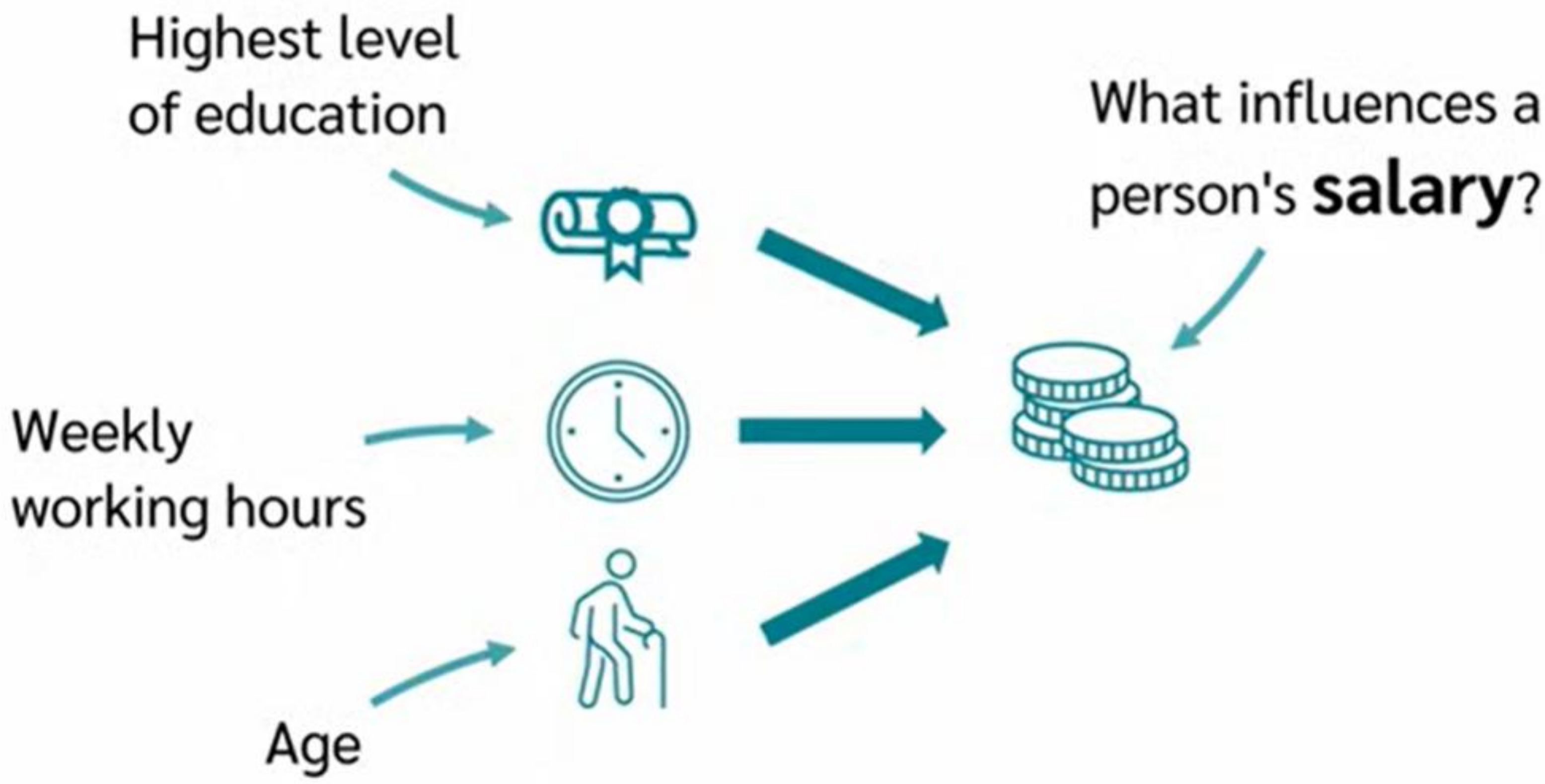
$$y = 108.35 \cdot \mathbf{3} + 60919.44 = 61244.49$$

$$y = 108.35 \cdot \mathbf{4} + 60919.44 = 61352.84$$

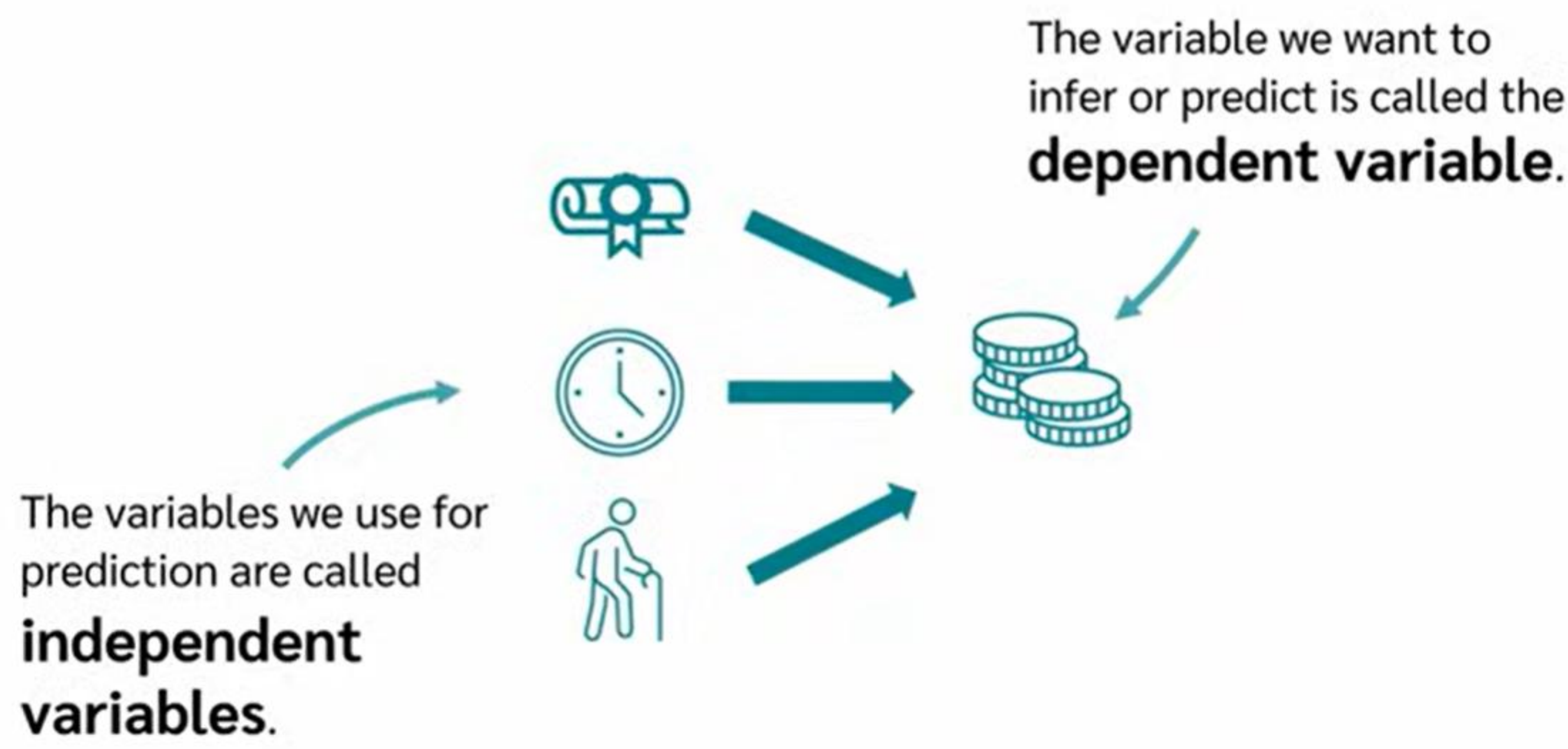
Multiple Linear Regression?



Multiple Linear Regression?



Multiple Linear Regression?



Multiple Linear Regression?

In the case of **linear regression**,
this was our **equation:**

$$y = b \cdot x + a$$

One ***dependent variable***

One ***independent variable***

In **multiple linear regression**, we have
more than one independent variable.

$$y = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

Multiple Linear Regression?

$$y = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

The **coefficients b** and **a**
are interpreted similar to those in
simple linear regression.

Let's make one
small adjustment

$$\boxed{\hat{y}} = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

Multiple Linear Regression?

$$y = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

The **coefficients b** and **a**
are interpreted similar to those in
simple linear regression.

Let's make one
small adjustment

$$\boxed{\hat{y}} = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

Represents the
predicted values
from the regression model,

while ***y*** denotes the
observed, actual
values.

Multiple Linear Regression?

1. *What are the Assumptions?*
2. *How to calculate a regression?*
3. *How to interpret the results?*
4. *What are Dummy Variables?*

Multiple Linear Regression?

1. What are the Assumptions?

   1. Linear Relationship
   2. Independence of Errors
   3. Homoscedasticity
   4. Normally Distributed Errors
   5. No multicollinearity

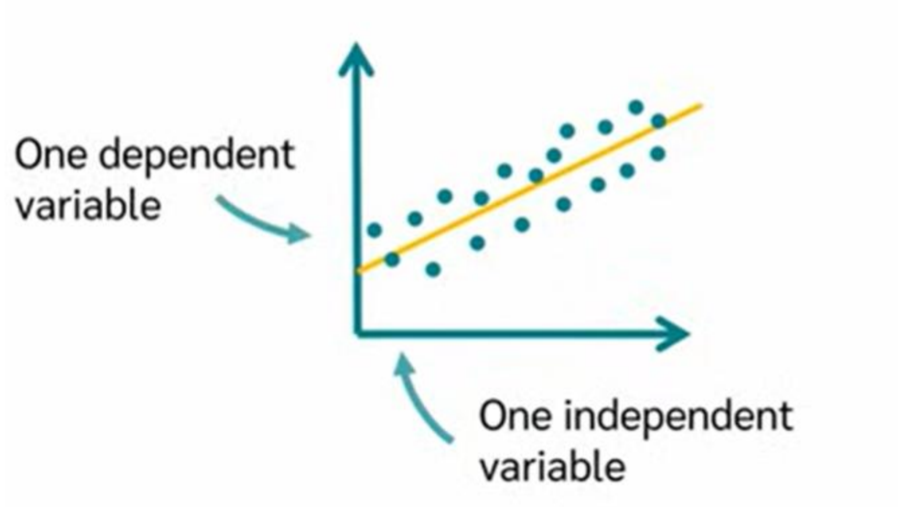
Multiple Linear Regression?

Linear Relationship

- A straight line is drawn through the data.
- This straight line should represent all points as good as possible.

Multiple Linear Regression?

Simple Linear Regression



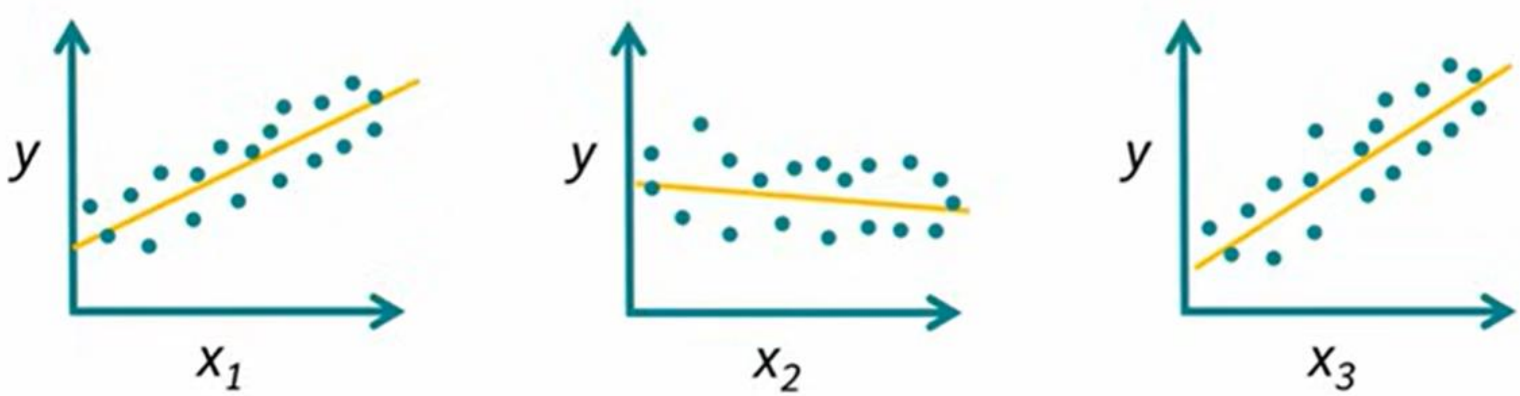
- **Multiple Linear Regression**

Multiple independent variables

$$y = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

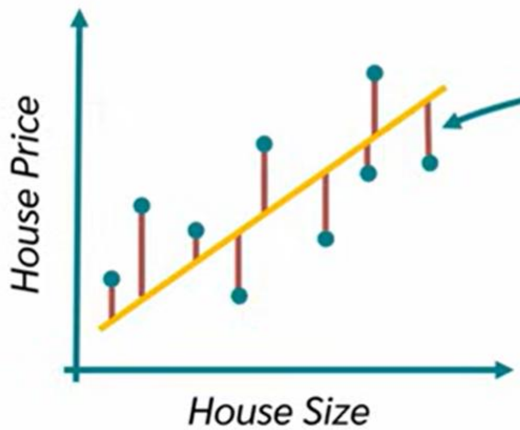
Multiple Linear Regression?

- **Multiple Linear Regression**
  - We can use each variable separately for visualization.
  - This is used for initial assessment whether the linear relationship exist.

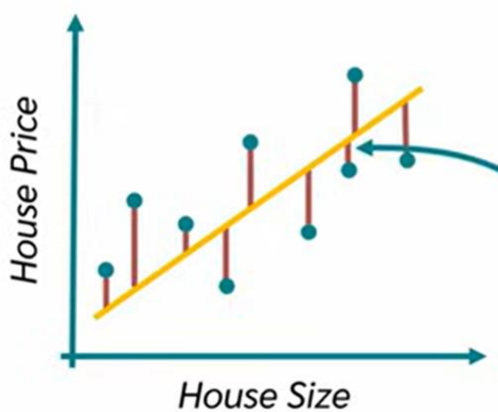


Multiple Linear Regression?

2. *Independence of Errors*



The errors should be independent of each other.



This means that the error of one point doesn't affect another.

We can test this with the **Durbin-Watson Test.**

Durbin-Watson-Test

Copy

Autocorrelation	Statistics	p
-0.16	2.19	.616

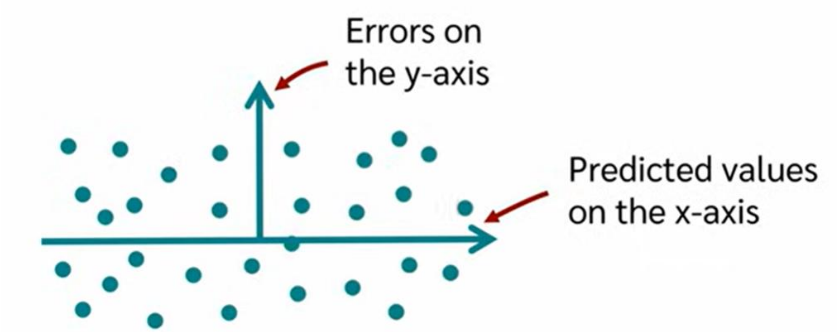
Used to detect **autocorrelation (correlation between errors)**

Interpretation:
- Value ≈ **2** → No autocorrelation (good)
- Value < **2** → Positive autocorrelation
- Value > **2** → Negative autocorrelation

Multiple Linear Regression?

3. *Homoscedasticity*

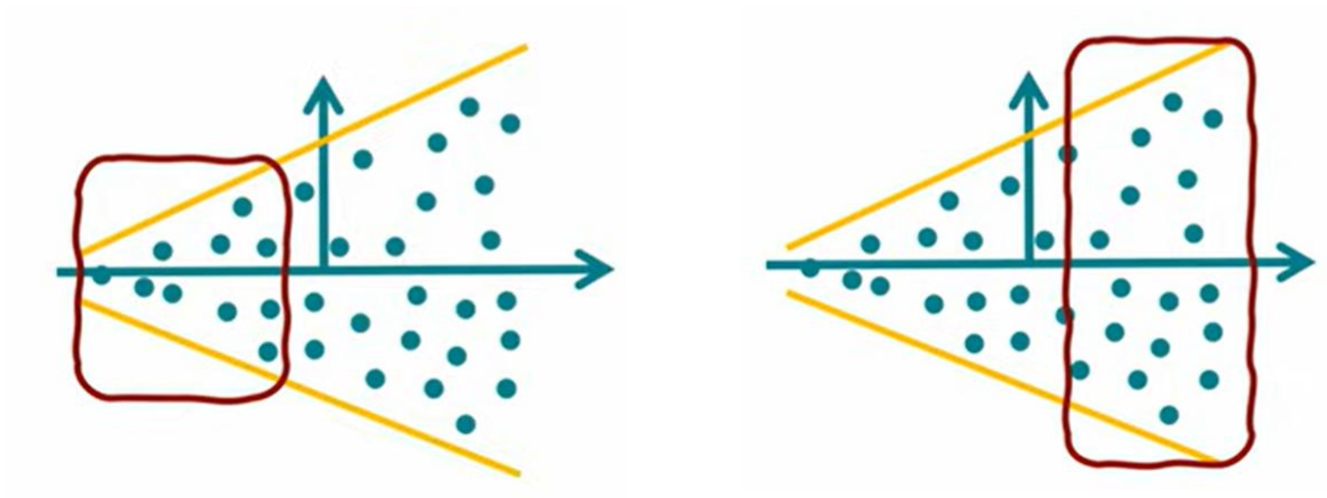
Homoscedasticity, or homogeneity of variances, occurs in statistics when the error terms (residuals) are consistent and have equal variance across all levels of an independent variable in a regression model.



- If we plot the predicted values of x-axis and errors on y-axis, there spread should be roughly the same.
- The variance should be equal.

Multiple Linear Regression?

3. *Homoscedasticity*

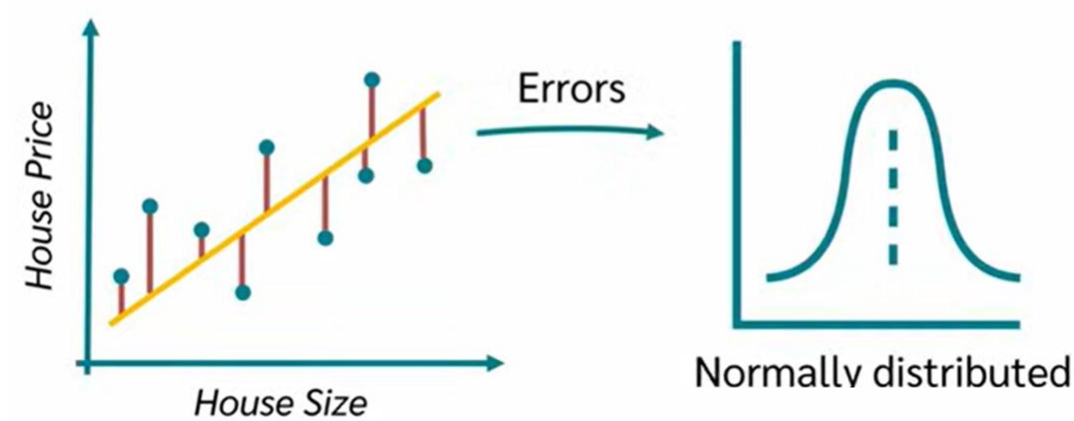


- In the above figure the error changes with the values of x-axis.
- The variance changes when the x values are increased.

Multiple Linear Regression?

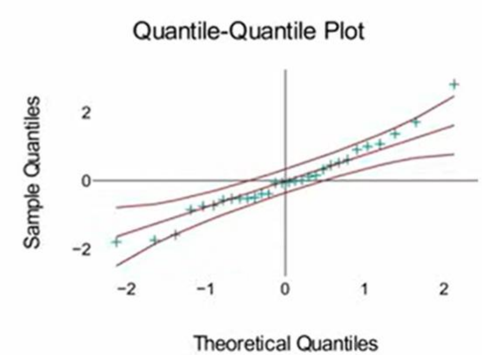
4. Normally Distributed Errors

Normally distributed errors mean that our model makes small, random mistakes that follow a bell-shaped pattern. We verify this using a QQ-plot—if points lie on a straight line, our assumption is satisfied.



- The errors should be normally distributed.

We can test this with a **QQ-Plot**



Multiple Linear Regression?

What is

Multicollinearity

Multicollinearity means that **two or more** independent variables

$$\hat{y} = b_1 \cdot \boxed{x_1} + b_2 \cdot \boxed{x_2} + \ldots + b_k \cdot x_k + a$$

are **highly correlated** with each other.

The effect of individual variables **cannot** be clearly **separated**.

$$\hat{y} = b_1 \cdot \boxed{x_1} + b_2 \cdot \boxed{x_2} + \ldots + b_k \cdot \boxed{x_k} + a$$

Multiple Linear Regression?

- **Why is this the problem?**

The effect of individual variables **cannot** be clearly **separated**.

$$\hat{y} = b_1 \cdot \boxed{x_1} + b_2 \cdot \boxed{x_2} + \ldots + b_k \cdot \boxed{x_k} + a$$

- If two variables are highly correlated

Highly correlated

$$\hat{y} = b_1 \cdot \boxed{x_1} + b_2 \cdot \boxed{x_2} + \ldots + b_k \cdot x_k + a$$

- We cannot find the effect of b1 and b2.

$$\hat{y} = \boxed{b_1} \cdot x_1 + \boxed{b_2} \cdot x_2 + \ldots + b_k \cdot x_k + a$$

Multiple Linear Regression?

If both are
completely equal,

$$\hat{y} = b_1 \cdot \boxed{x_1} + b_2 \cdot \boxed{x_2} + \ldots + b_k \cdot x_k + a$$

$$\hat{y} = \boxed{b_1} \cdot x_1 + \boxed{b_2} \cdot x_2 + \ldots + b_k \cdot x_k + a$$

the **regression model**
cannot determine how large $\boldsymbol{b_1}$
and how large $\boldsymbol{b_2}$ should be.

This means that one
independent variable

can be **predicted** from
the **others** with a high
degree of accuracy.

$$\hat{y} = b_1 \cdot \boxed{x_1} + b_2 \cdot \boxed{x_2} + \ldots + b_k \cdot \boxed{x_k} + a$$

Multiple Linear Regression?

Example

$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

Price of house | Size of house | Number of rooms | other variables

Larger houses... *...tend to have* ***more rooms.***

$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

Price of house | Size of house | Number of rooms | other variables

Correlation

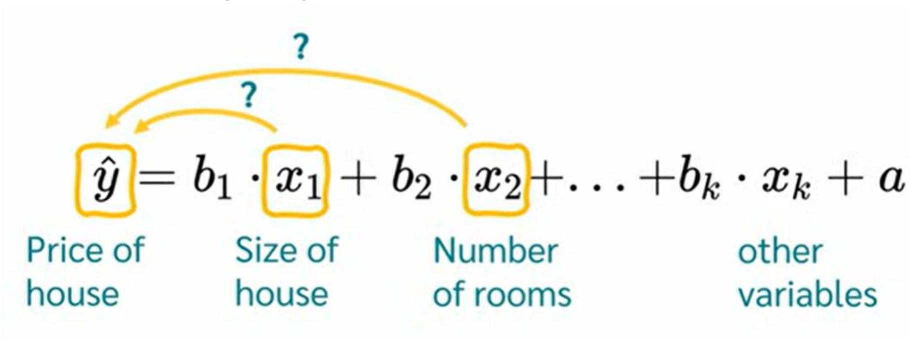
$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

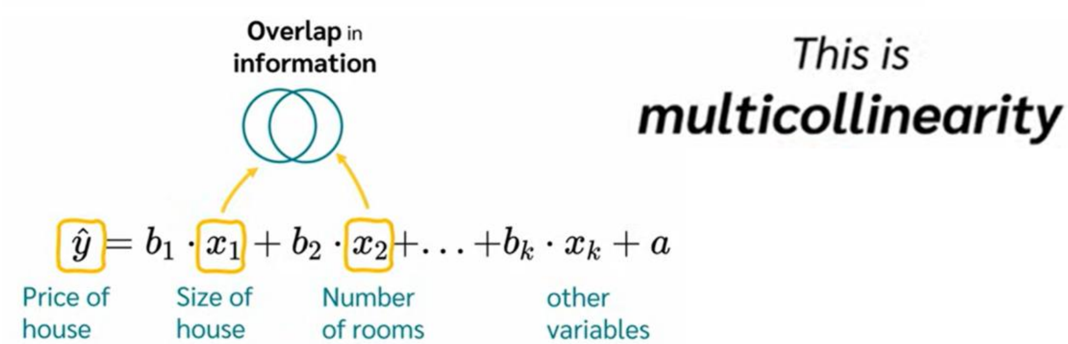
Price of house | Size of house | Number of rooms | other variables

Multiple Linear Regression?

Example

- If we include both x1 and x2, then model will struggle to analyze the effect of each variable if they are highly correlated to each other.





Multiple Linear Regression?

Example

- If the correlation is high, then the coefficient identification/computation is impossible to find.
- If we use regression model just for prediction, then multicollinearity is less critical.

Use the regression model
for a **prediction**

$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

Multicollinearity is
less critical!

Multiple Linear Regression?

Example

Calculate multiple linear regression

Interpret the results

Blood Pressure	Age	Weight	Cholesterol
78	58	76	153
91	48	70	234
79	34	52	195
93	62	99	190
...	...	...	...
...	...	...	...
...	...	...	...
90	40	55	223

Multiple Linear Regression?

Dependent Variable | *Independent Variable*

Blood Pressure	Age	Weight	Cholesterol
78	58	76	153
91	48	70	234
79	34	52	195
93	62	99	190
...	...	...	...
...	...	...	...
...	...	...	...
90	40	55	223

$$Blood\ Pressure = b_1 \cdot Age + b_2 \cdot Weight + b_3 \cdot Cholesterol + a$$

Multiple Linear Regression?

Coefficients

	Unstandardized Coefficients	Standardized Coefficients			
Model	B	Beta	Standard error	t	p
(Constant)	23.94		12.11	1.98	.059
Age	0.26	0.37	0.1	2.52	.018
Weight	0.25	0.4	0.09	2.89	.008
Cholesterol	0.16	0.51	0.05	3.43	.002

$$Blood\ Pressure = 0.26 \cdot Age + 0.25 \cdot Weight + 0.16 \cdot Cholesterol + 23.94$$

$$Blood\ Pressure = 0.26 \cdot Age + 0.25 \cdot Weight + 0.16 \cdot Cholesterol + 23.94$$

91 55 95 180

Multiple Linear Regression?

What if the independent variables are nominal?

Independent variables

$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

Nominal

But how do we use **nominal variables** in a regression model as **independent variables**?

$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k + a$$

Multiple Linear Regression?

Category coded with 0 is called reference category.

Reference category

1 = male

0 = female

Gender

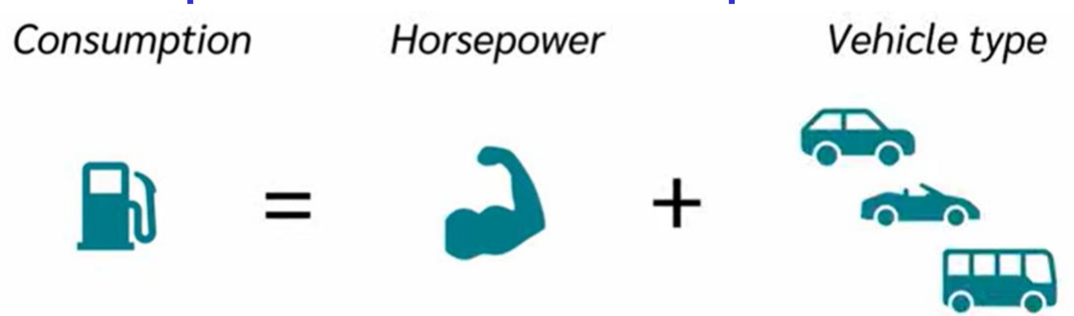
1 = male

0 = female

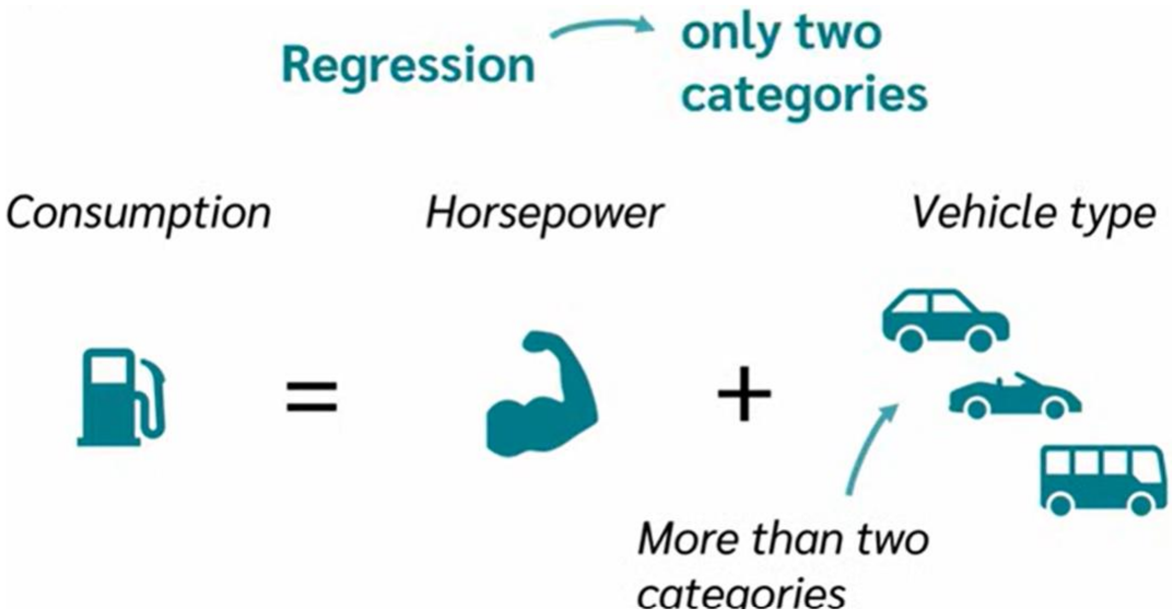
Multiple Linear Regression?

Variables having values more than 2.

If we want to predict the consumption of fuel?

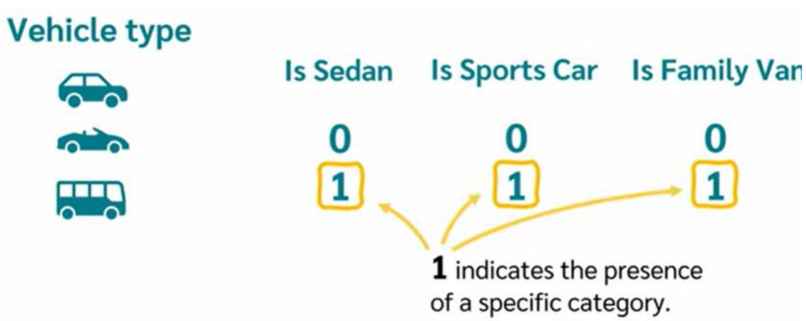


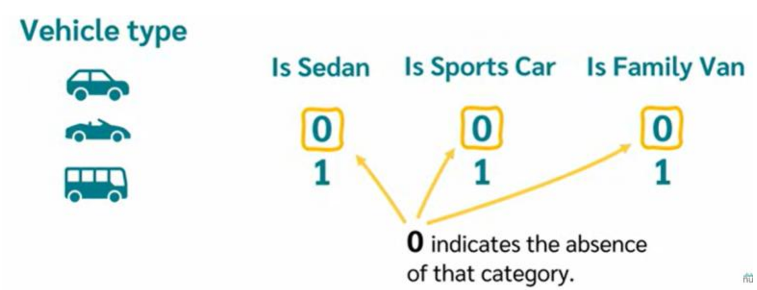
Multiple Linear Regression?



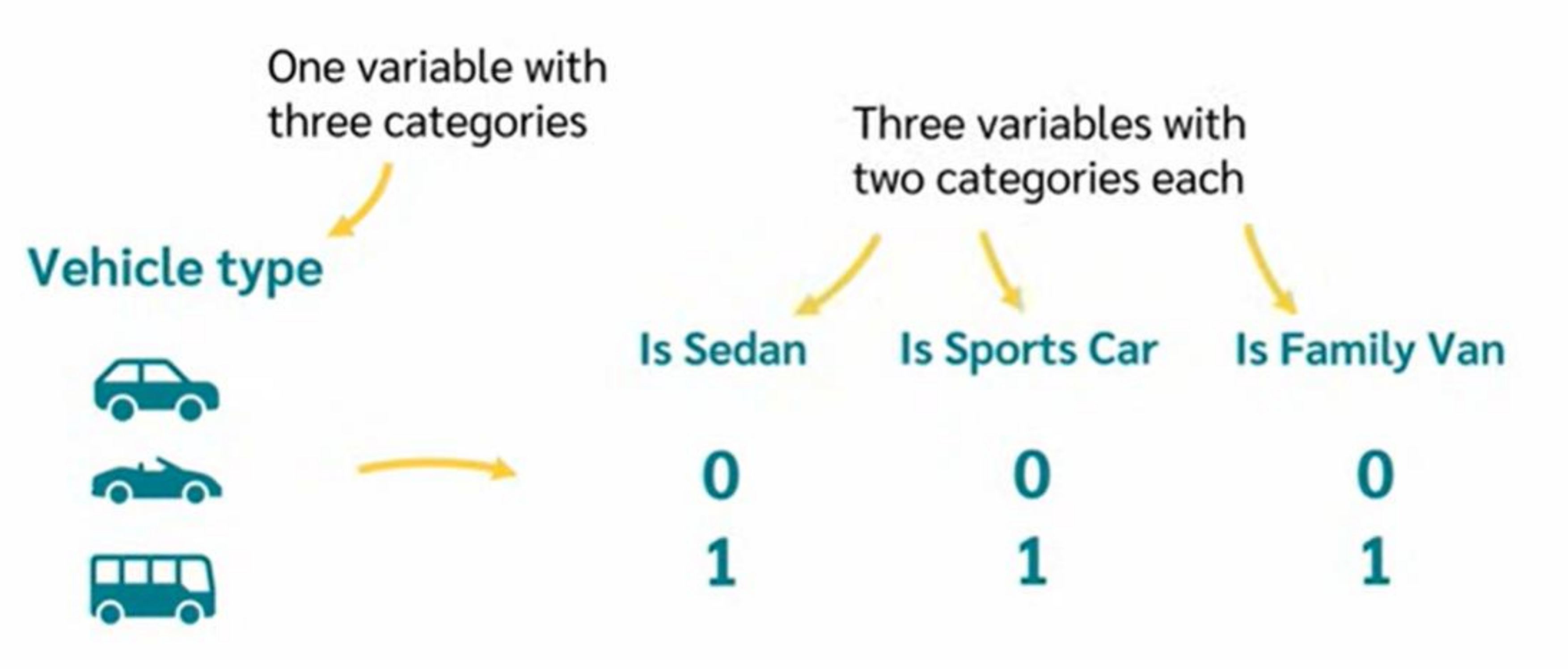
Dummy variables are **artificial variables** that make it possible to handle variables with **more than two categories**.

Multiple Linear Regression?





Multiple Linear Regression?



Multiple Linear Regression?

Vehicle type	IS Sedan	IS Sports Car	IS Family Van
Sedan	1	0	0
Sedan	1	0	0
Sports car	0	1	0
Van	0	0	1
Sports car	0	1	0
...	...	...	...

Vehicle type	IS Sedan	IS Sports Car	IS Family Van
Sedan	1	0	0
Sedan	1	0	0
Sports car	0	1	0
Van	0	0	1
Sports car	0	1	0
...	...	...	...

Multiple Linear Regression?

One important thing to note!

The **number** of dummy variables you create will always be the number of **categories minus one**.

Three categories

Only two dummy variables

Is Sedan	Is Sports Car	Is Family Van
1 ✓	0 ✗	0 ✗
1	0	0
0 ✗	1 ✓	0 ✗
0 ✗	0 ✗	1 ✓
0	1	0
...	...	...

Conclusion

Regression is fundamental in ML

Used for smoothing, prediction, classification